

Course Outcomes (COs)	Course Outcome Statements
CO1	Analyse DC circuits using fundamental circuit laws to determine voltage, current, power, and energy.
CO2	Analyse AC circuits using phasor representation, impedance, resonance, and power factor and modern power system concepts
CO3	Apply magnetic circuit concepts for Electrical equipment and machines
CO4	Assess the operational behaviour of a single-phase transformers to justify their selection for voltage control in everyday power systems.
CO5	Assess the operational behaviour of AC/DC and special electrical machines to justify their selection for industrial applications.

Mapped CO(s) Number	Relevant Major Theory Session Outcomes (TSOs)	Relevant Units/ Body of Knowledge (BoK)
CO1	<i>TSO 1a.</i> Define basic electrical quantities: charge, current, voltage, resistance, inductance, capacitance and EMF. <i>TSO 1b.</i> Apply Ohm's Law to simple resistive circuits <i>TSO 1c.</i> Analyze effect of temperature on resistance <i>TSO 1d.</i> Apply Kirchhoff's Laws in circuit analysis <i>TSO 1e.</i> Solve DC networks to determine power and energy.	Unit- 1.0: Electrical fundamentals and DC Circuits (8 hrs) 1.1 Concept of Charge, Current, Voltage, Resistance, Inductance and Capacitance, EMF 1.2 Ohm's Law and Kirchhoff's Current and Voltage Laws 1.3 Effect of Temperature on Resistance 1.4 Resistor, Inductor & Capacitor: Properties and Applications 1.5 Series, parallel and series parallel resistor circuits, 1.6 Current and Voltage Divider Rules 1.7 Star-Delta and Delta-star transformations
CO2	<i>TSO 2a.</i> Represent sinusoidal quantities using phasors. <i>TSO 2b.</i> Calculate impedance of R, L, C, RL, RC and RLC circuits. <i>TSO 2c.</i> Analyze series and parallel resonance; determine resonant frequency, Q-factor, and bandwidth. <i>TSO 2d.</i> Calculate real, reactive, apparent power and power factor. <i>TSO 2e.</i> Explain the benefits of improving power factor	Unit- 2.0: AC Circuits (10 hrs) 2.1 Generation of sinusoidal EMF; Peak, RMS, average values; form factor; Wave and phasor representation 2.2 Impedance of R, L, C circuits; RL, RC, RLC series and parallel circuits; series and parallel resonance. 2.3 Power in AC circuits: Active, reactive, apparent power 2.4 Power factor - Unity, lag, lead, Power factor improvement, advantages 2.5 Harmonics, source and its impact

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	<i>TSO 2f.</i> Explain the source of harmonics and its impact. <i>TSO 2g.</i> Explain the need and advantages of three-phase systems <i>TSO 2h.</i> Evaluate line and phase relationships in balanced systems <i>TSO 2i.</i> Measure and compute three-phase power in balanced circuits	2.6 Three-phase systems, concept need and advantages 2.7 Single Phase vs Three Phase System (Advantages), Phase sequence 2.8 Balanced and unbalanced supply and load and line-phase relationships 2.9 Voltage, Current relationship and power in Star and Delta connections 2.10 Power in balanced three phase circuits
CO3	<i>TSO 3a.</i> Define various terms related to magnetic circuits <i>TSO 3b.</i> Explain the basic properties of the given Magnetic materials. <i>TSO 3c.</i> Interpret the B-H curve and hysteresis loop of the given material. <i>TSO 3d.</i> Analyze analogy with electric circuits <i>TSO 3e.</i> Evaluate magnetic losses <i>TSO 3f.</i> Perform magnetic circuit calculations <i>TSO 3g.</i> Explain the principles of electromagnetic induction with the help of Faraday's laws <i>TSO 3h.</i> and concepts of self and mutual inductance. <i>TSO 3i.</i> Justify the application of Fleming's Right-hand rule and Fleming's Left-hand rule to DC Generator and DC motor	Unit- 3.0: Magnetic Circuits and Electromagnetic induction (10 hrs) 3.1 Magnetic circuits - Magnetic Flux, Magnetic intensity, Flux Density(B), Magnetomotive Force (MMF) 3.2 Reluctance, Permeability, simple numerical 3.3 Analogy between Electric and Magnetic circuits 3.4 Basic Properties of Magnetic materials 3.5 Magnetic Circuits with DC Excitation 3.6 Magnetic Leakage 3.7 B-H Curve 3.8 Hysteresis and Eddy Current Losses 3.9 Magnetic Circuit Calculations 3.10 Electromagnetic induction - Faradays Laws, Fleming's right hand and left-hand Rules, 3.11 Statically and dynamically induced EMF; Concepts of self-inductance, mutual inductance and coefficient of coupling; Energy stored in magnetic fields.
CO4	<i>TSO 4a.</i> Describe the construction, working principle of a single-phase transformer. <i>TSO 4b.</i> Evaluate transformer performance using load test <i>TSO 4c.</i> Differentiate between power and distribution transformer <i>TSO 4d.</i> List the special purpose transformer and their uses.	Unit- 4.0: Transformer (8 hrs) 4.1 Transformer - Construction, principle of operation, Single phase v/s three phase transformer 4.2 Emf equation 4.3 Transformer performance – Load test -Voltage regulation, losses and efficiency. 4.4 Single phase v/s 3 phase transformer and its uses 4.5 Special purpose transformers- Introduction and applications only- Audio transformer, Radio Frequency (RF) Transformer, Intermediate Frequency (IF) transformer, Pulse transformer,

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		Isolation transformer, Impedance matching transformer
CO5	<i>TSO 5a.</i> Describe the parts and materials used in a DC machine. <i>TSO 5b.</i> Justify the use of different DC motors for a specific application <i>TSO 5c.</i> Explain concept of rotating magnetic field, slip and need of starters for three-phase induction motors. <i>TSO 5d.</i> Describe the starting methods of a single-phase induction motor. <i>TSO 5e.</i> Describe the construction, working and applications of a given special electrical machine.	Unit- 5.0: Electrical and Special Purpose Machines (6 hrs) 5.1 DC machines: construction, EMF equation, types of DC Motor and applications, need of DC motor starter 5.2 AC machines: Three-phase induction motor construction, working, types, slip, need of starter, DOL, star-delta starter 5.3 Construction, working principle of single-phase induction motors, Starting methods 5.4 Construction, working and applications of - (i) BLDC motor (ii) Servo motor (iii) Stepper motor (iv) Switched Reluctance motor (v) Universal motor (vi) Coreless motor

Text / References: -

S. No.	Titles	Author(s)	Publisher and Edition with ISBN
1.	Basic Electrical and Electronics Engineering	Vinoth Kumar K., Saravanakumar R., Jegathesan V., Kowsalya M., Mahesh M.	Wiley India, New Delhi, 2nd Ed., 2022, ISBN: 978-9354643194
2.	Fundamentals of Electrical and Electronics Engineering (with Lab Manual)	Susan S. Mathew and Saji T. Chacko	Khanna Book Publishing Co., 2021, ISBN: 978-9391505592
3.	Basic Electrical Engineering	V. Mittle, Arvind Mittle	McGraw Hill Education, 2017 ISBN: 978-0070593572
4.	Basic Electrical Engineering	D.C. Kulshreshtha	Tata McGraw-Hill, 3rd Ed., 2019, ISBN: 978-0070083929
5.	Basic Electrical and Electronics Engineering	D.P. Kothari & I.J. Nagrath	Tata McGraw Hill, 2nd Ed., 2020 ISBN: 978-9389811247
6.	Principles of Electrical Engineering	V.K. Mehta and Rohit Mehta	S. Chand & Co., 2nd Ed., 2018,

S. No.	Titles	Author(s)	Publisher and Edition with ISBN
			ISBN: 978-8121924870
7.	Basic Electrical and Electronics Engineering	M.S. Sukhija, T. K. Nagsarkar	Oxford University press,2012 ISBN: 978-0198081807
8.	A Course in Electrical and Electronic Measurements and Instrumentation	A.K. Sawhney and Puneet Sawhney	Dhanpat Rai & Co., 19th Ed., 2018, ISBN: 978-8177001006
9.	Basic Electrical Engineering.	C.L. Wadhwa	New Age International, 2007 ISBN-10: 9788122421521 ISBN-13: 978-8122421521
10.	Textbook of Electrical Technology – Volume I	B.L. Theraja& A.K Theraja,	S. Chand Publication, 2022, 23 edition, ISBN-10: 8121924405 ISBN-13: 978-8121924405
11.	Textbook of Electrical Technology AC and DC Machines –Volume II	B.L. Theraja& A.K Theraja,	S. Chand Publication,2025 ISBN 10:9355018258 ISBN 13:9789355018250